Lecture 10 Review Handout

Statistics 367-1-4361

Lecture Overview

Lecture 10 introduces model comparison as a way to choose among plausible statistical models rather than only rejecting or accepting a null model. The lecture begins with overfitting and underfitting, then reframes model quality around scientific goals: theory building, hypothesis testing, and prediction. The main technical path is predictive model comparison: KL divergence motivates the score, information criteria approximate it, and cross-validation estimates it more directly. The lecture ends by placing model comparison inside the Bayesian workflow and returning to the birth-weight and smoking example.

Lecture Sections

10.1 — Overfitting and underfitting

This section starts from the problem that many models are possible for the same dataset. Measures such as R^2 reward fit to the observed data, but fit alone always improves as parameters are added. The goal is not maximal fit; it is useful complexity, balancing models that ignore structure against models that chase noise.

New Definitions and Terms

- **Overfitting** — fitting details of the observed data that do not generalize to new data
- **Underfitting** — using a model too simple to capture important structure in the data
- **Goodness of fit** — how closely a model fits the observed data
- **Occam's razor** — the preference for the simpler model when two models fit equally well
- **Chatton's anti-razor** — the reminder that added assumptions may be necessary when simpler assumptions do not explain the phenomena

Figure

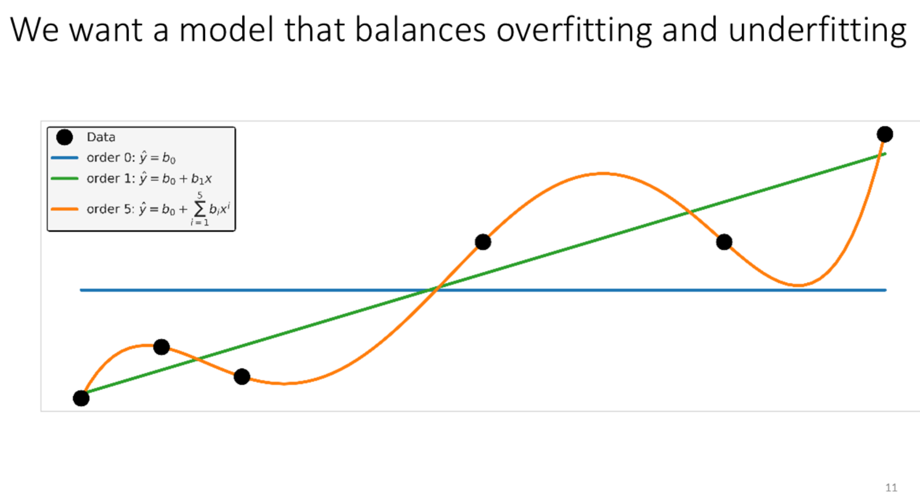


Figure 1: Slide 11 - A polynomial model balancing overfitting and underfitting

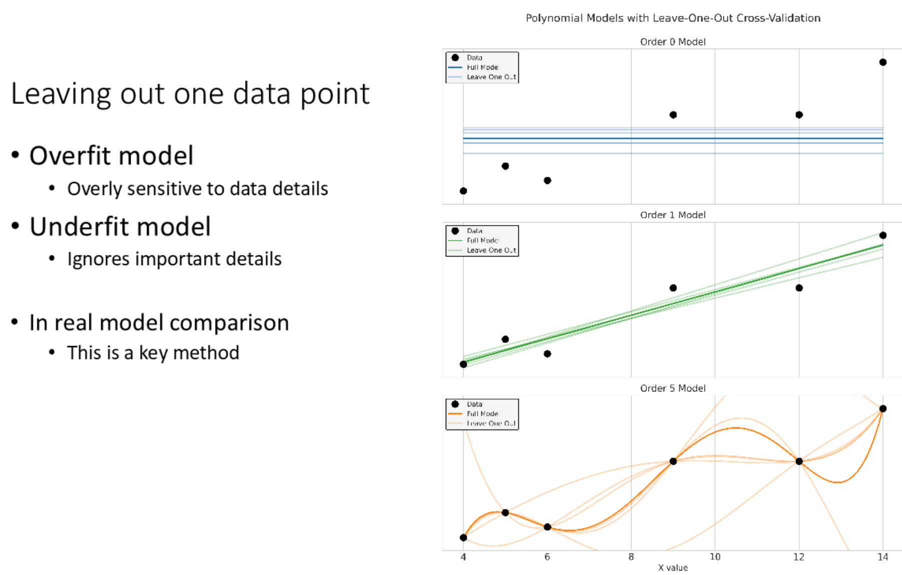


Figure 2: Slide 14 — Leaving out one data point

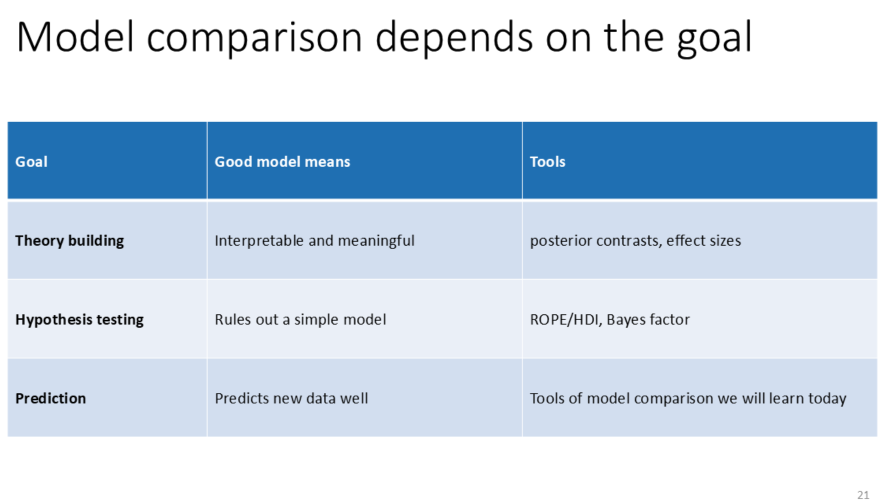
10.2 — What makes a good model

This section makes model evaluation explicitly goal-dependent. A model can be useful for theory building, hypothesis testing, or prediction, and each goal implies a different meaning of "good." The central shift is from asking whether a model is true to asking whether it is useful for the scientific goal.

New Definitions and Terms

- **Theory-building model** — a model evaluated by whether it is interpretable, meaningful, and generalizable
- **Predictive model** — a model evaluated by how well it predicts new data
- **Candidate model** — one of the models being compared for a particular scientific purpose

Figure



Goal	Good model means	Tools
Theory building	Interpretable and meaningful	posterior contrasts, effect sizes
Hypothesis testing	Rules out a simple model	ROPE/HDI, Bayes factor
Prediction	Predicts new data well	Tools of model comparison we will learn today

Figure 3: Slide 21 — Model comparison depends on the goal

10.3 — Kullback-Leibler divergence

This section motivates predictive comparison by imagining a distance between a candidate model and the true data-generating distribution. We do not know the true distribution, so we cannot calculate this distance directly. The key insight is that a model closer to the true data-generating distribution should assign higher probability to new data.

New Definitions and Terms

- **Surprise** — the information cost of an outcome; lower-probability outcomes are more surprising
- **Entropy** — the expected surprise under the true distribution
- **Cross-entropy** — the expected surprise when guesses are made using the wrong distribution
- **Kullback-Leibler divergence** — the extra expected surprise from using one distribution in place of another
- **Score** — the expected log predictive density of new data under a model
- **Deviance** — -2 times the score; the same predictive idea expressed on a different scale

Formulas

Entropy:

$$H(p) = -\int p(y) \log p(y) \, dy$$

Cross-entropy:

$$H(p, q) = -\int p(y) \log q(y) \, dy$$

KL divergence:

$$D_{\mathrm{KL}}(p \| q) = H(p, q) - H(p) = \int p(y) \log \frac{p(y)}{q(y)} \, dy$$

For model comparison, the wrong distribution is the model's posterior predictive distribution:

$$D_{\mathrm{KL}}\{p \| p(y \mid m)\} = \int p(y) \{\log p(y) - \log p(y \mid m)\} \, dy$$

The predictive score is the part that can differ between models:

$$\text{score}(m) \approx \sum_i \log p(\tilde{y}_i \mid m)$$

where

$$p(\tilde{y}_i \mid m) = \int p(\tilde{y}_i \mid \theta, m) p(\theta \mid m)\, d\theta$$

Deviance uses the same comparison on a different scale:

$$\text{deviance} = -2\,\text{score}$$

Figure

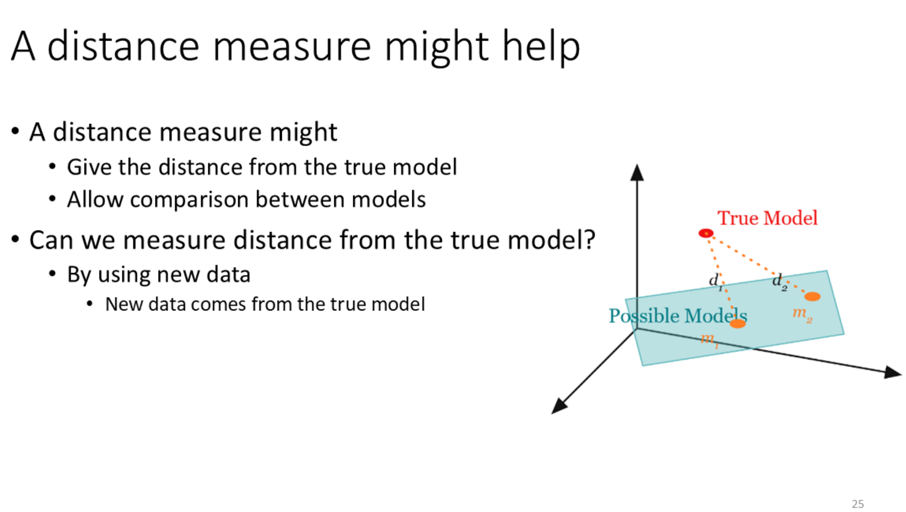


Figure 4: Slide 25 — A distance measure might help

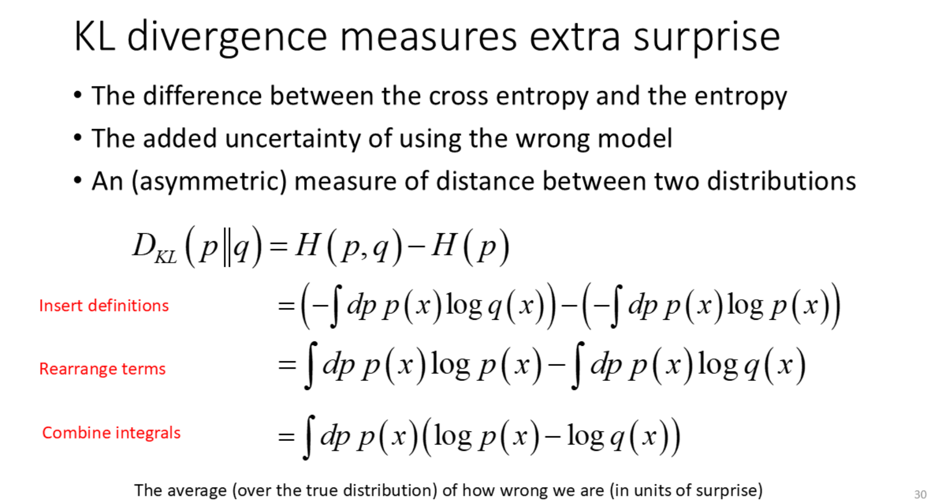


Figure 5: Slide 30 — KL divergence measures extra surprise

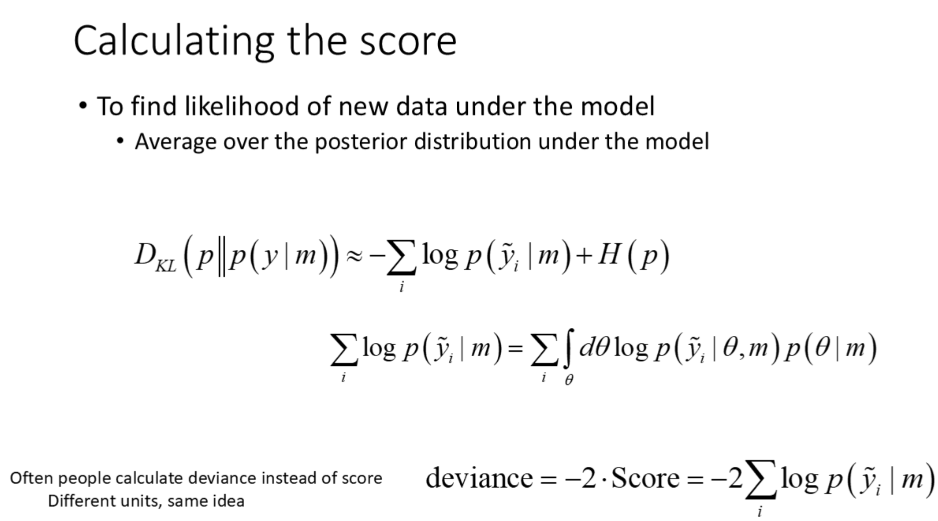


Figure 6: Slide 33 — Calculating the score

10.4 — Information criteria

Information criteria estimate out-of-sample predictive performance using only the observed data. The problem is optimism: a model usually predicts the data it was fit on better than it predicts new data. Different information criteria estimate and subtract this optimism in different ways.

New Definitions and Terms

- **Information criterion** — an estimate of out-of-sample predictive performance based on in-sample fit plus an optimism correction
- **Optimism** — the excess predictive performance obtained by testing a model on the same data used to fit it
- **AIC** — Akaike information criterion; a frequentist information criterion using a point estimate and a parameter-count penalty
- **WAIC** — widely applicable information criterion; a Bayesian information criterion using the full posterior distribution
- **Effective number of parameters** — the posterior-based model-complexity estimate used by WAIC
- **Model weight** — a weight for combining model predictions according to estimated predictive performance

Formulas

AIC:

$$\mathrm{AIC} = -2 \log p(y \mid \hat{\theta}, m) + 2k$$

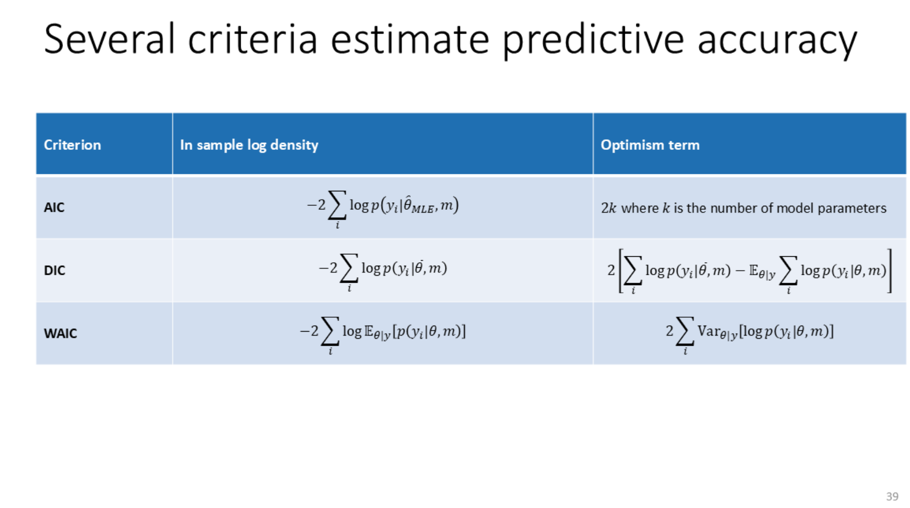
Here k is the number of model parameters. Lower AIC means better estimated predictive performance.

WAIC:

$$\mathrm{WAIC} = -2 \sum_i \log \int p(y_i \mid \theta, m) p(\theta \mid m)\, d\theta + 2 \sum_i \mathrm{Var}_\theta \{\log p(y_i \mid \theta, m)\}$$

The first term measures posterior predictive fit to the observed data. The second term estimates optimism using the posterior variation in pointwise log likelihood.

Figure



Several criteria estimate predictive accuracy

Criterion	In sample log density	Optimism term
AIC	$-2\sum_i \log p(y_i \mid \hat{\theta}_{MLE}, m)$	$2k$ where k is the number of model parameters
DIC	$-2\sum_i \log p(y_i \mid \bar{\theta}, m)$	$2\left[\sum_i \log p(y_i \mid \bar{\theta}, m) - \mathbb{E}_{\theta \mid y} \sum_i \log p(y_i \mid \theta, m)\right]$
WAIC	$-2\sum_i \log \mathbb{E}_{\theta \mid y}[p(y_i \mid \theta, m)]$	$2\sum_i \mathrm{Var}_{\theta \mid y}[\log p(y_i \mid \theta, m)]$

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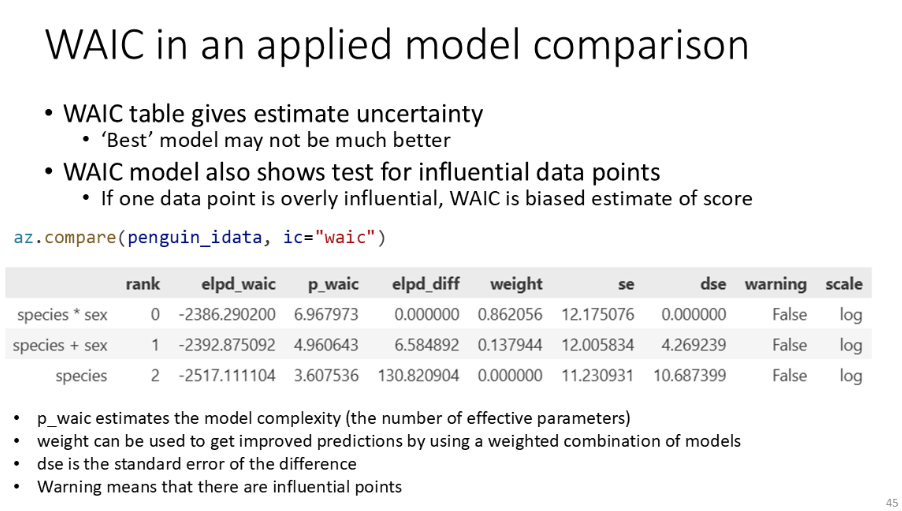
Figure 7: Slide 39 — Several criteria estimate predictive accuracy



WAIC in an applied model comparison

- WAIC table gives estimate uncertainty
  - ‘Best’ model may not be much better
- WAIC model also shows test for influential data points
  - If one data point is overly influential, WAIC is biased estimate of score

```
az.compare(penguin_idata, ic="waic")
```

	rank	elpd_waic	p_waic	elpd_diff	weight	se	dse	warning	scale
species * sex	0	-2386.290200	6.967973	0.000000	0.862056	12.175076	0.000000	False	log
species + sex	1	-2392.875092	4.960643	6.584892	0.137944	12.005834	4.269239	False	log
species	2	-2517.111104	3.607536	130.820904	0.000000	11.230931	10.687399	False	log

- p_waic estimates the model complexity (the number of effective parameters)
- weight can be used to get improved predictions by using a weighted combination of models
- dse is the standard error of the difference
- Warning means that there are influential points

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Figure 8: Slide 45 — WAIC in an applied model comparison

10.5 — Cross-validation

Cross-validation estimates predictive performance by repeatedly fitting a model on one part of the data and evaluating it on another part. True validation would use a new dataset, but that is often expensive or slow. Leave-one-out cross-validation is especially attractive in Bayesian modeling because it can often be approximated without refitting the model N times.

New Definitions and Terms

- **Validation** — testing a model on data not used to fit it
- **True validation** — validation on a newly collected dataset
- **Withheld data** — data held out from fitting and used for testing
- **Leakage** — information from the test data influencing fitting or model selection
- **Cross-validation** — repeated train-test splitting so each subset is used for evaluation
- **K-fold cross-validation** — cross-validation in which the data are divided into k chunks
- **Leave-one-out cross-validation** — cross-validation in which each observation is left out once
- **ELPD** — expected log predictive density; the expected log likelihood of new data
- **Importance sampling** — estimating an expectation under one distribution using samples from another distribution and weights
- **PSIS** — Pareto-smoothed importance sampling; a stabilized approximation used for LOO-CV
- **Pareto k** — a diagnostic for the reliability of the PSIS approximation

Formulas

Expected log predictive density for LOO-CV:

$$\mathrm{ELPD}_{\mathrm{LOO-CV}} = \sum_i \log \int p(y_i \mid \theta) p(\theta \mid y_{-i})\, d\theta$$

The importance-sampling approximation uses posterior draws from $p(\theta \mid y)$ instead of refitting with y_i left out:

$$\mathrm{ELPD}_{\mathrm{LOO-CV}} \approx \sum_i \log \frac{\frac{1}{S}\sum_s w_s p(y_i \mid \theta_s)}{\frac{1}{S}\sum_s w_s}$$

with

$$w_s = \frac{1}{p(y_i \mid \theta_s)}$$

The Pareto k diagnostic checks whether the importance weights are too extreme. If $k > 1$, the LOO-CV estimate may be unreliable and full cross-validation may be required.

Figure

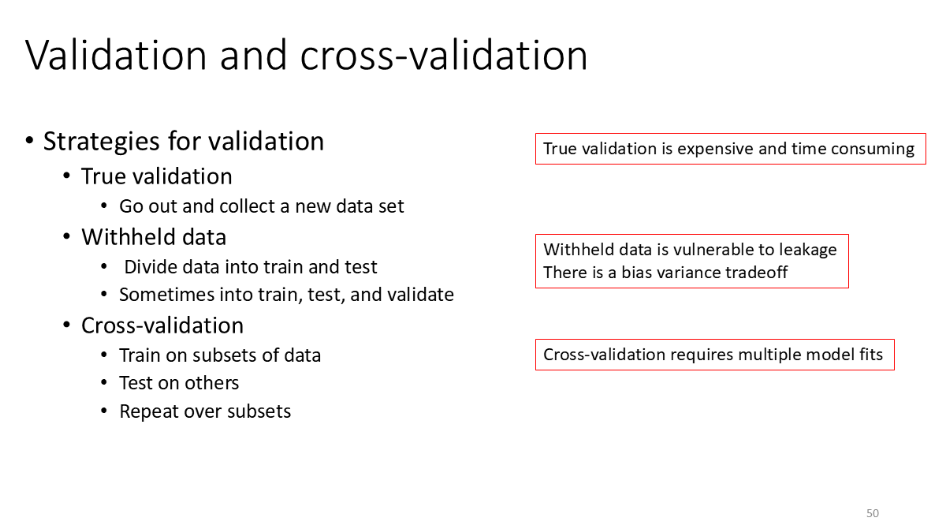

Figure 9: Slide 50 — Validation and cross-validation

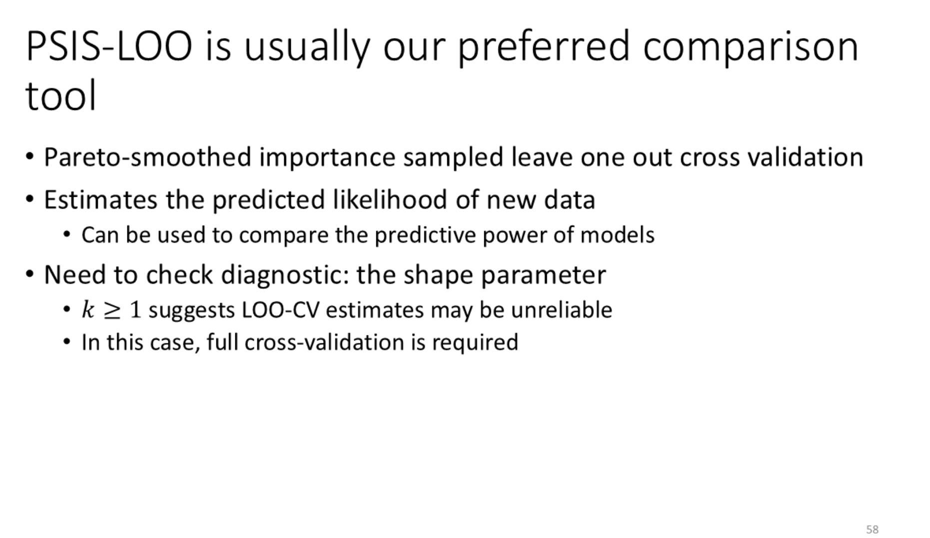

Figure 10: Slide 58 — PSIS-LOO is usually our preferred comparison tool

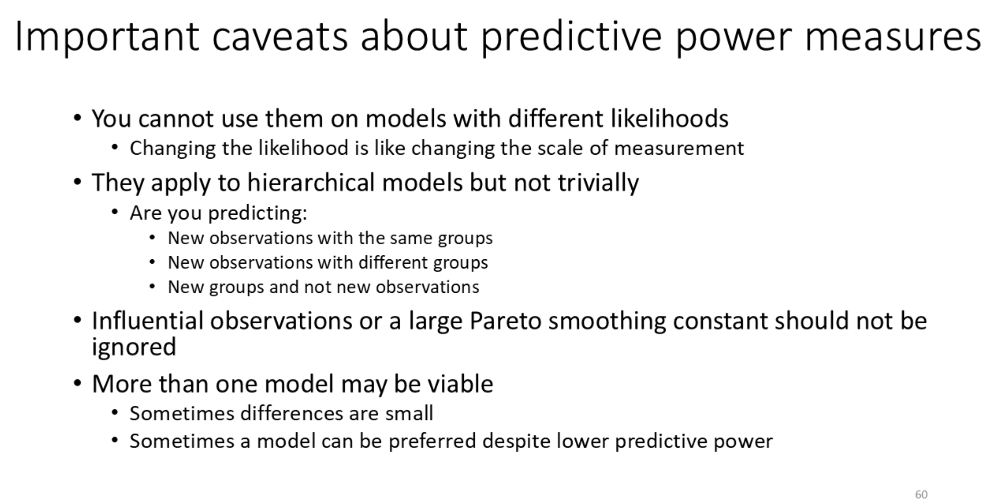

Figure 11: Slide 60 — Important caveats about predictive power measures

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10.6 — Updated Bayesian workflow

This section inserts model comparison into the full Bayesian workflow. Predictive comparison comes after scientific questions, candidate models, prior predictive checks, sampling, diagnostics, and posterior predictive checks. The ranking of models by predictive power is useful, but it does not replace interpretation of effects or scientific judgment.

The birth-weight example shows why this matters. The candidate models have similar overall effect sizes and similar predictive power, while the smoking parameter remains small but reasonably consistent across the leading models. The interaction model gives the largest smoking effect but also the largest uncertainty, so the analysis does not provide strong evidence for a smoking-by-gestation interaction.

New Definitions and Terms

- **Predictive ranking** — ordering models by estimated out-of-sample predictive performance
- **Model comparison workflow** — the use of predictive comparison after model checking and before final interpretation

Figure

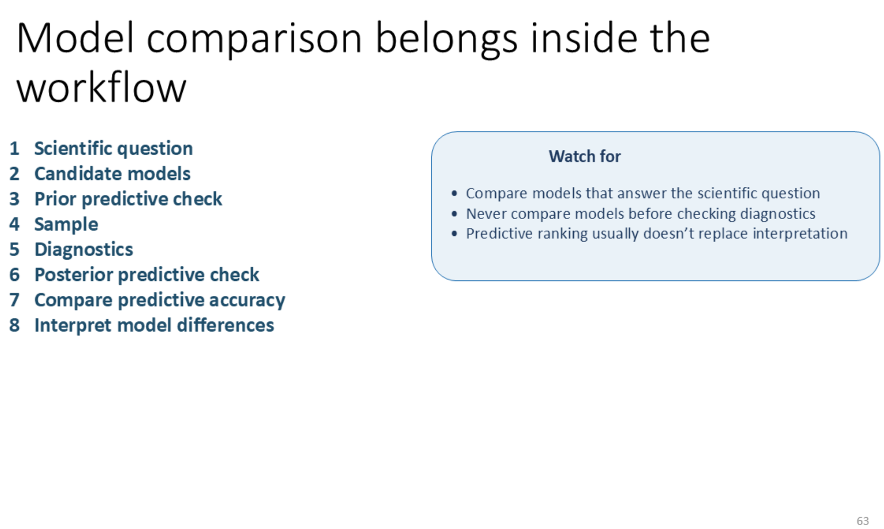

Figure 12: Slide 63 — Model comparison belongs inside the workflow

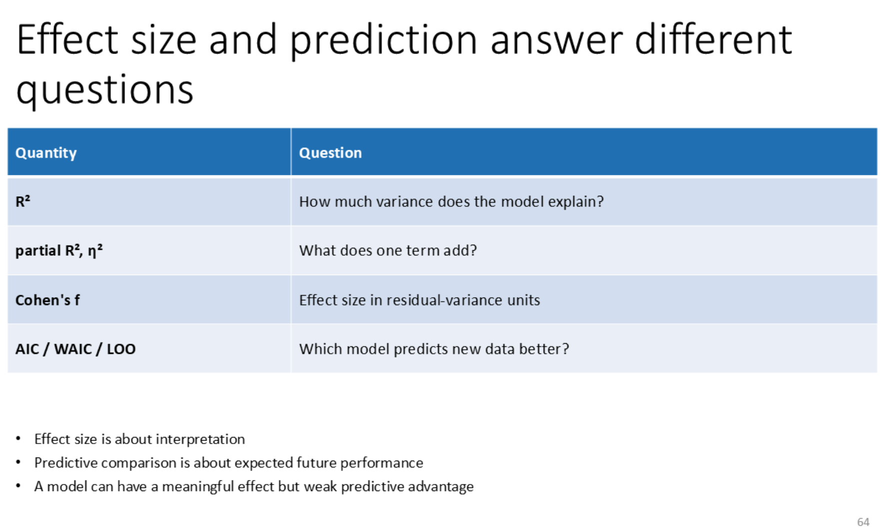

Figure 13: Slide 64 — Effect size and prediction answer different questions

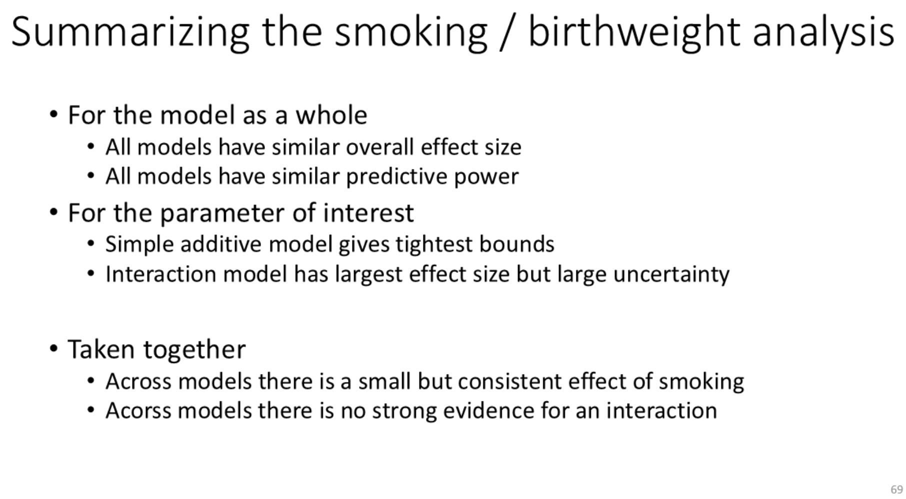

Figure 14: Slide 69 — Summarizing the smoking / birthweight analysis

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10.7 — Looking forward

The final slide previews the next step: combining regression with hierarchical models. The main takeaway from Lecture 10 is that model comparison is not an isolated calculation. It is one part of a larger workflow that starts with the scientific question and ends with interpreted model differences.

Figure

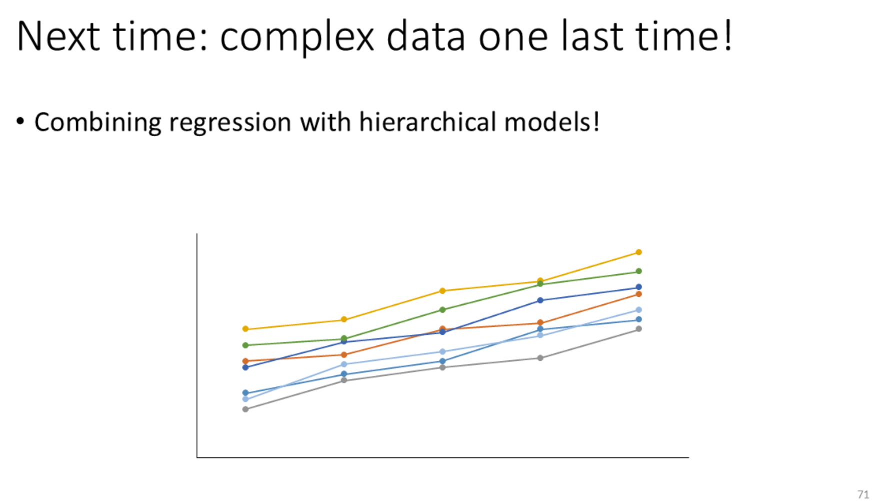

Figure 15: Slide 71 - Next time: combining regression with hierarchical models